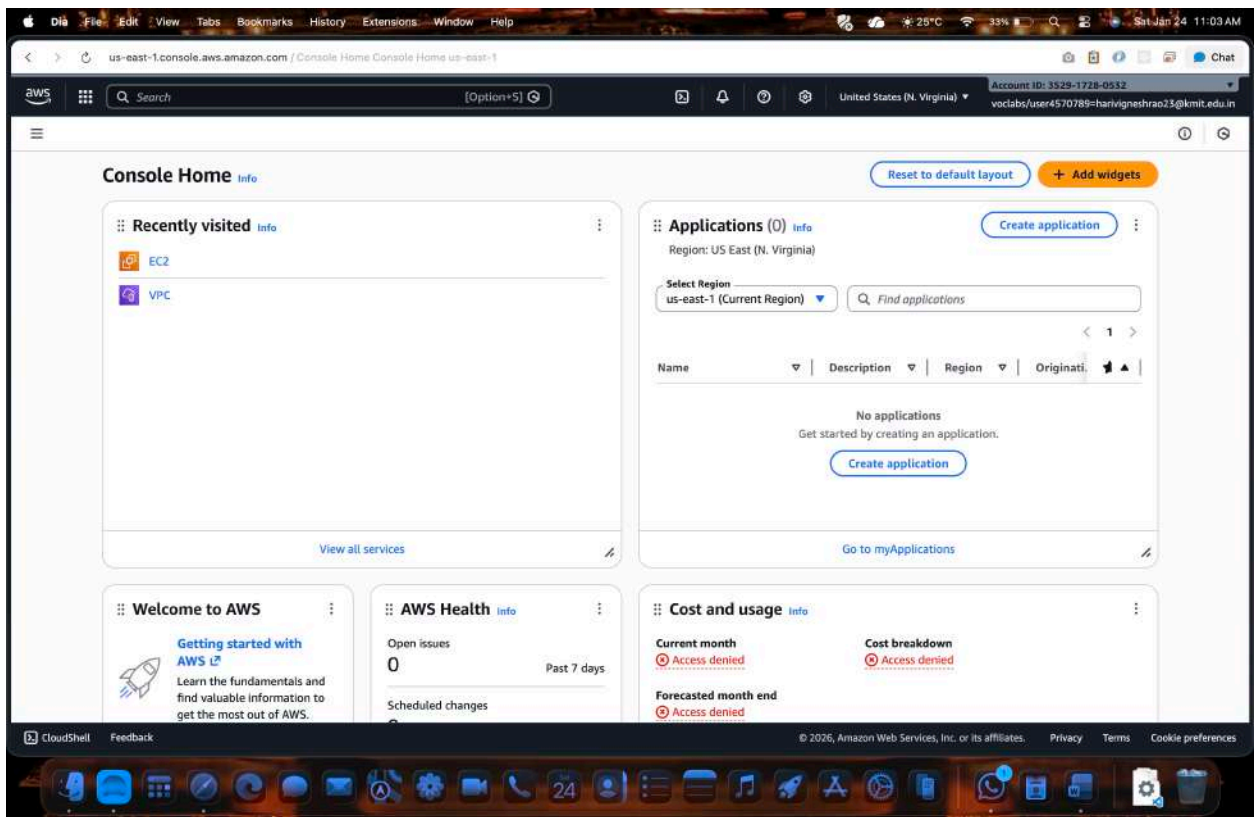


CLOUD COMPUTING LAB-3

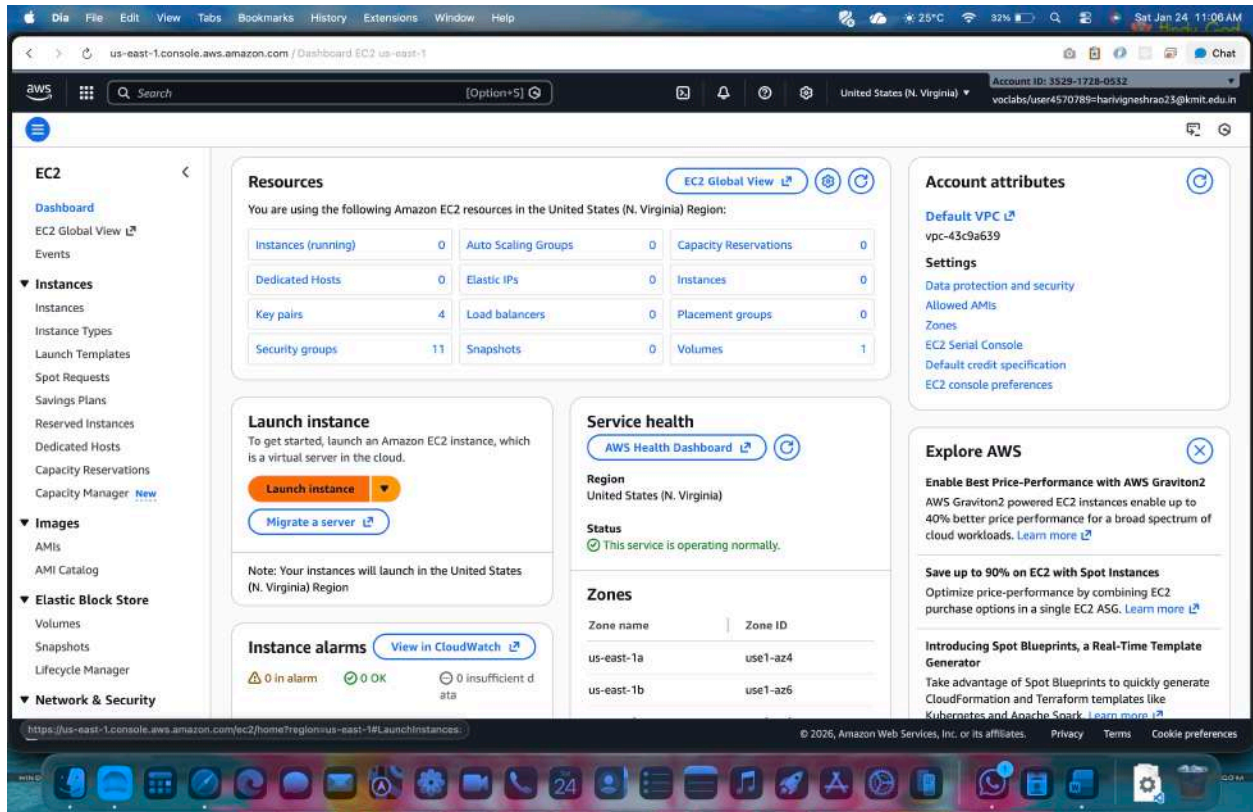
HARI VIGNESH RAO 23BD1A052Q

Step 1: Launching EC2 Instances

1. Sign in to AWS Management Console.



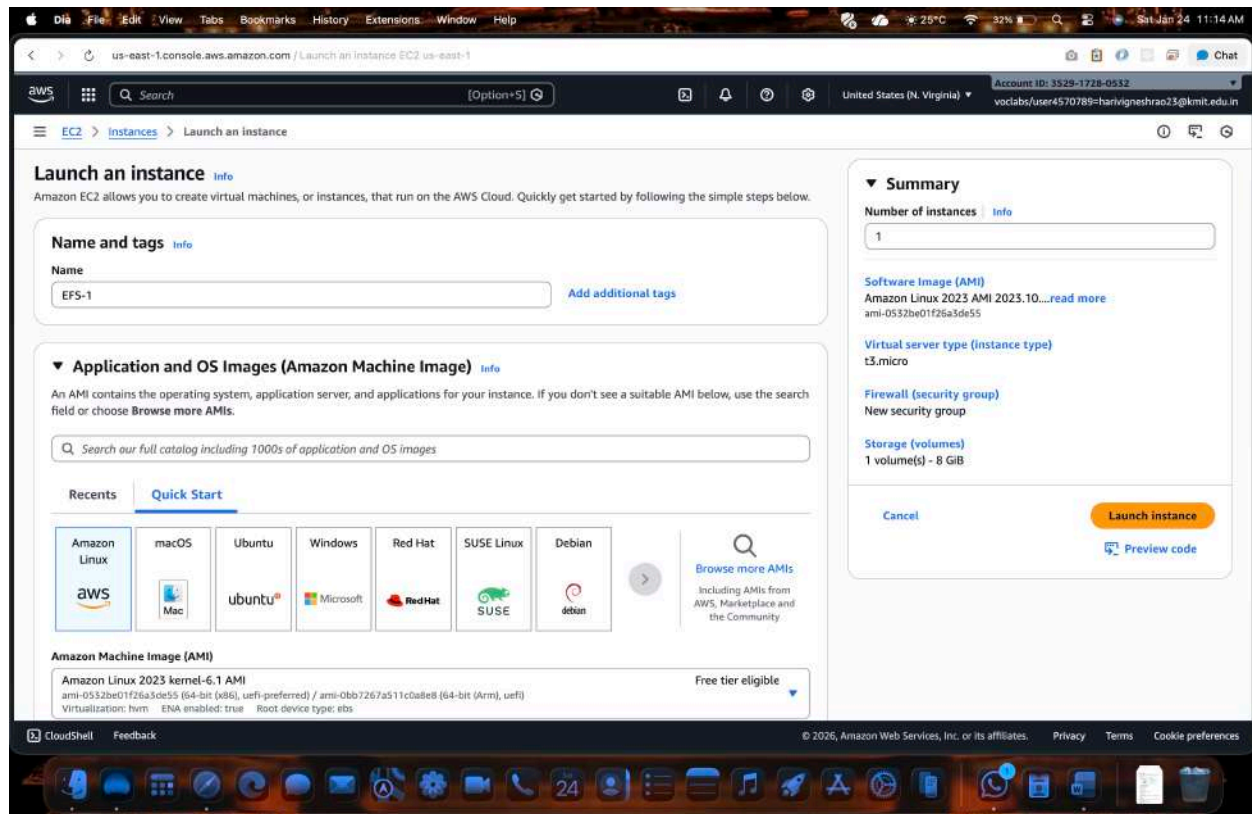
2. Go to the EC2 dashboard.
3. Click on "Launch Instance".



4. Configure instance details:

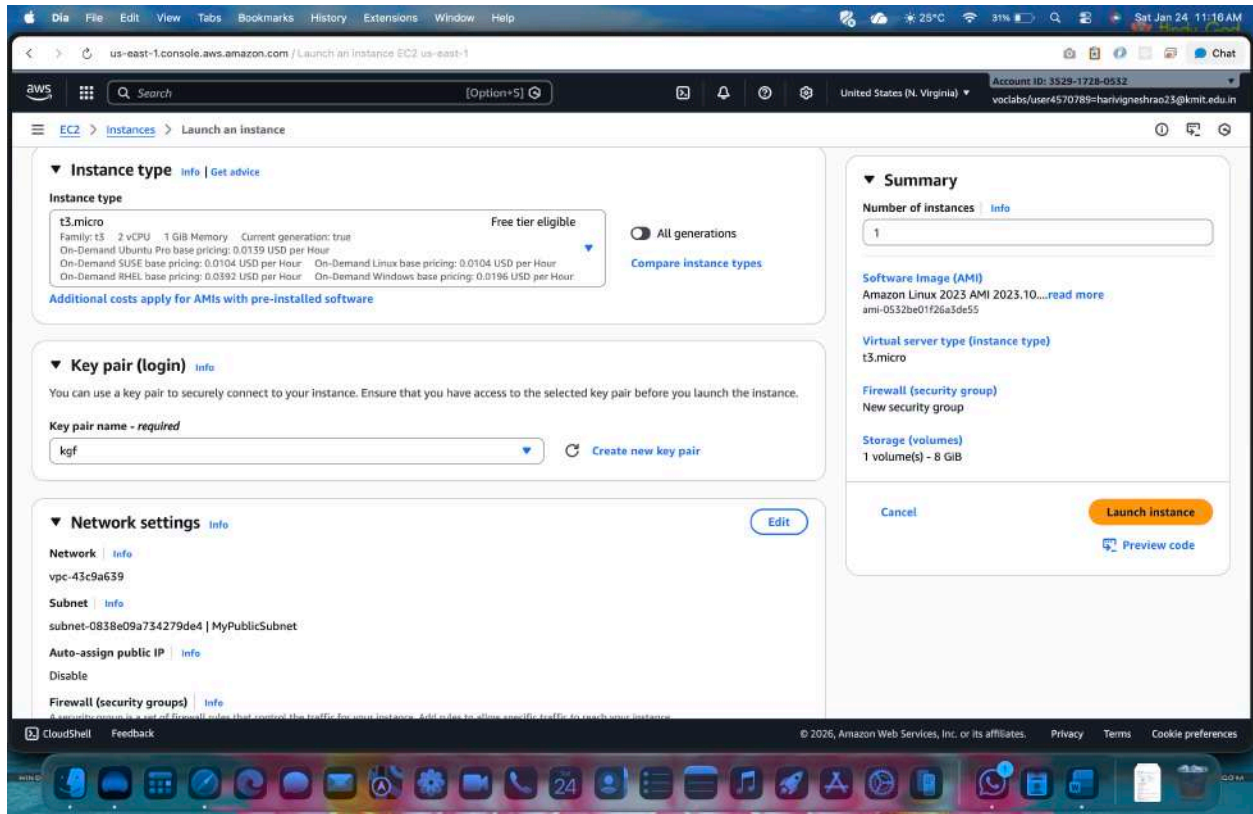
- Instance Name: **EFS-1**

5. Choose an Amazon Machine Image (AMI) for Linux.



6. Select instance type: t2.micro.

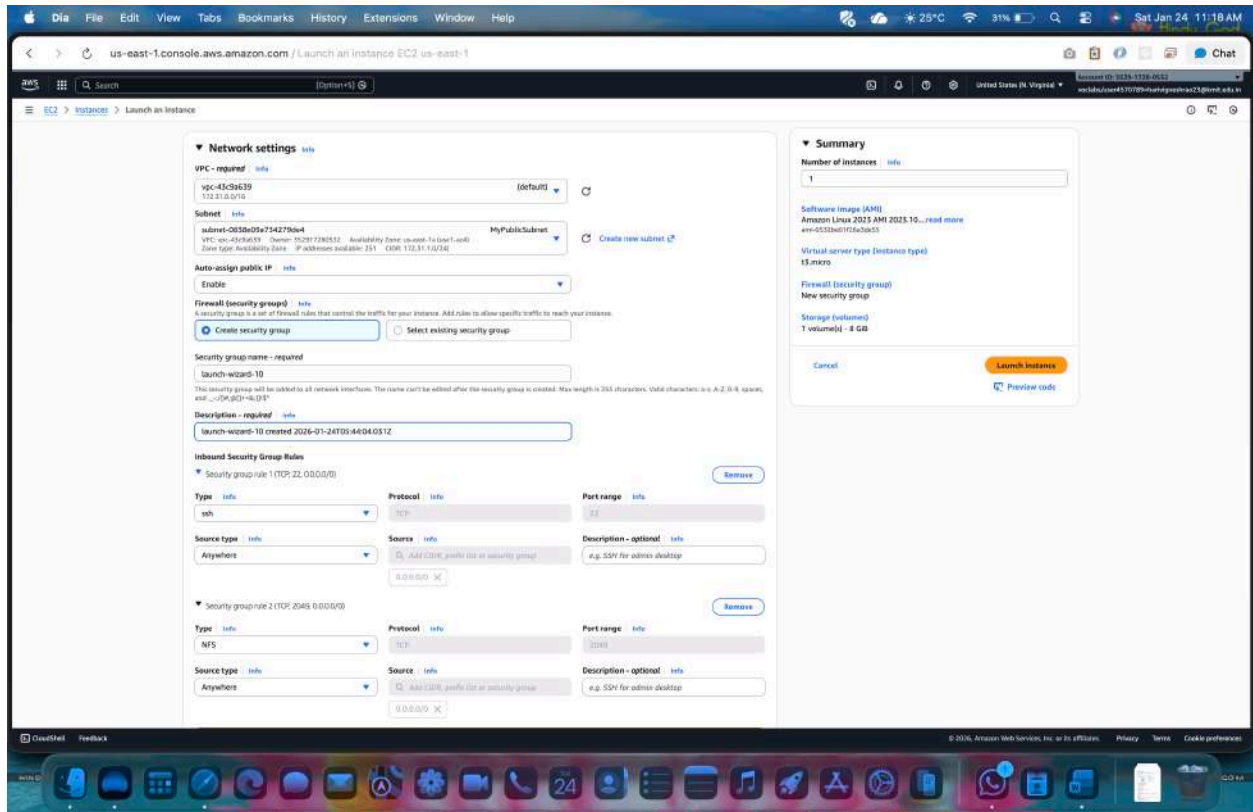
- Key Pair: kgf



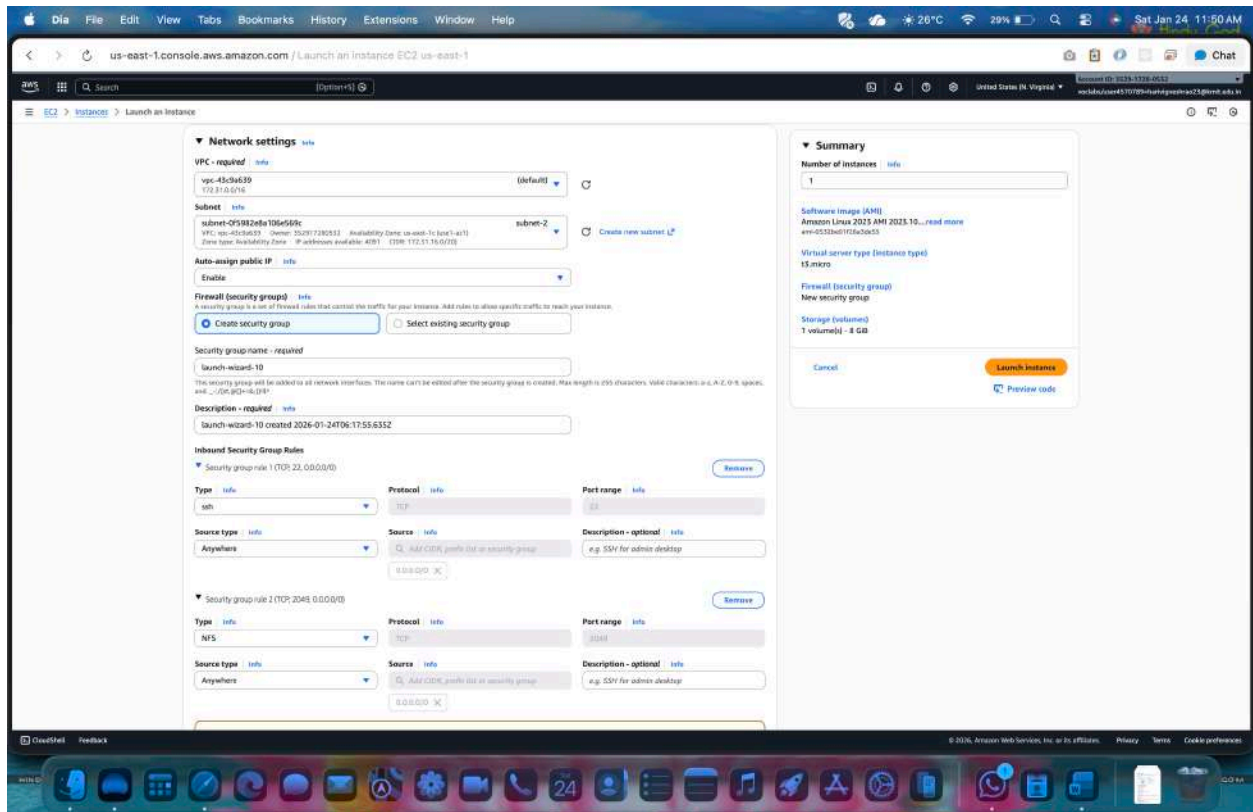
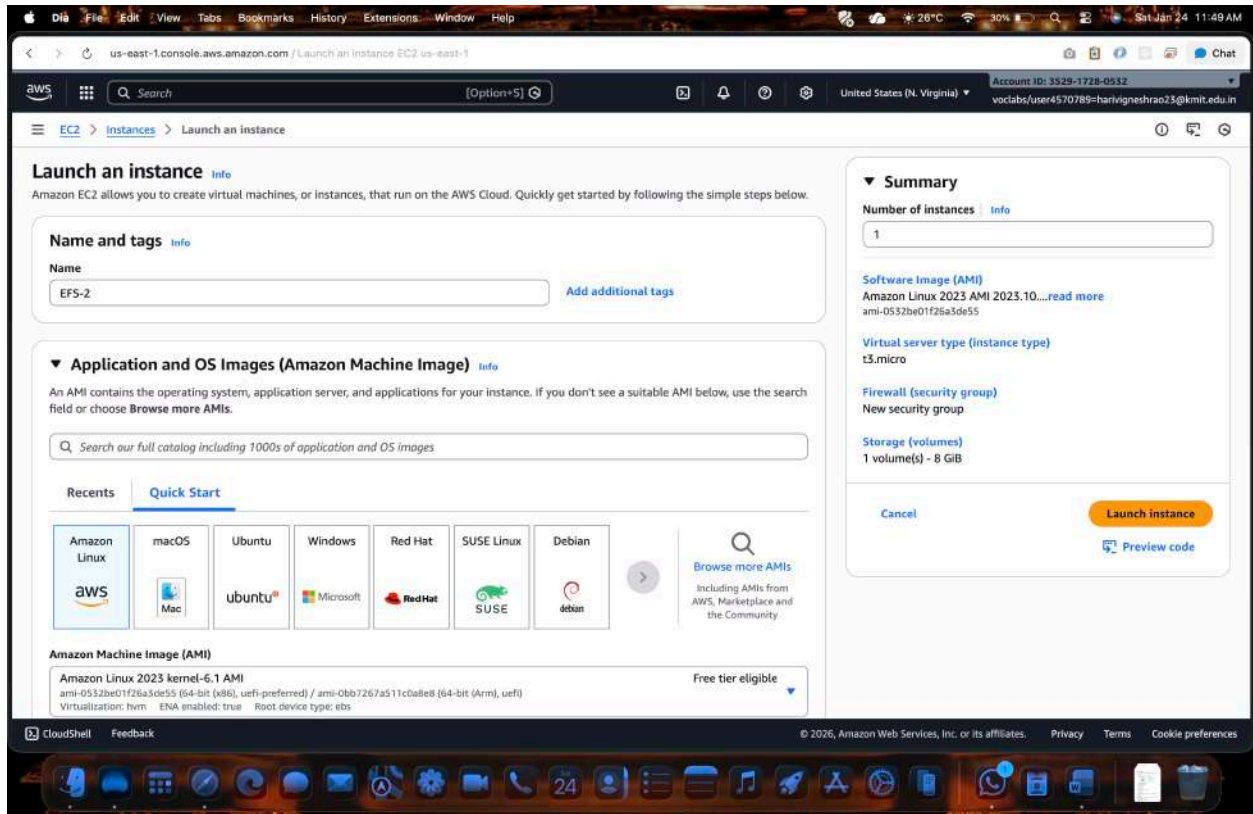
- **Network: Choose Subnet-1a**

- Security Group: **Create a new security group (SG) and add NFS and allow from anywhere.**

7. Launch the instance.

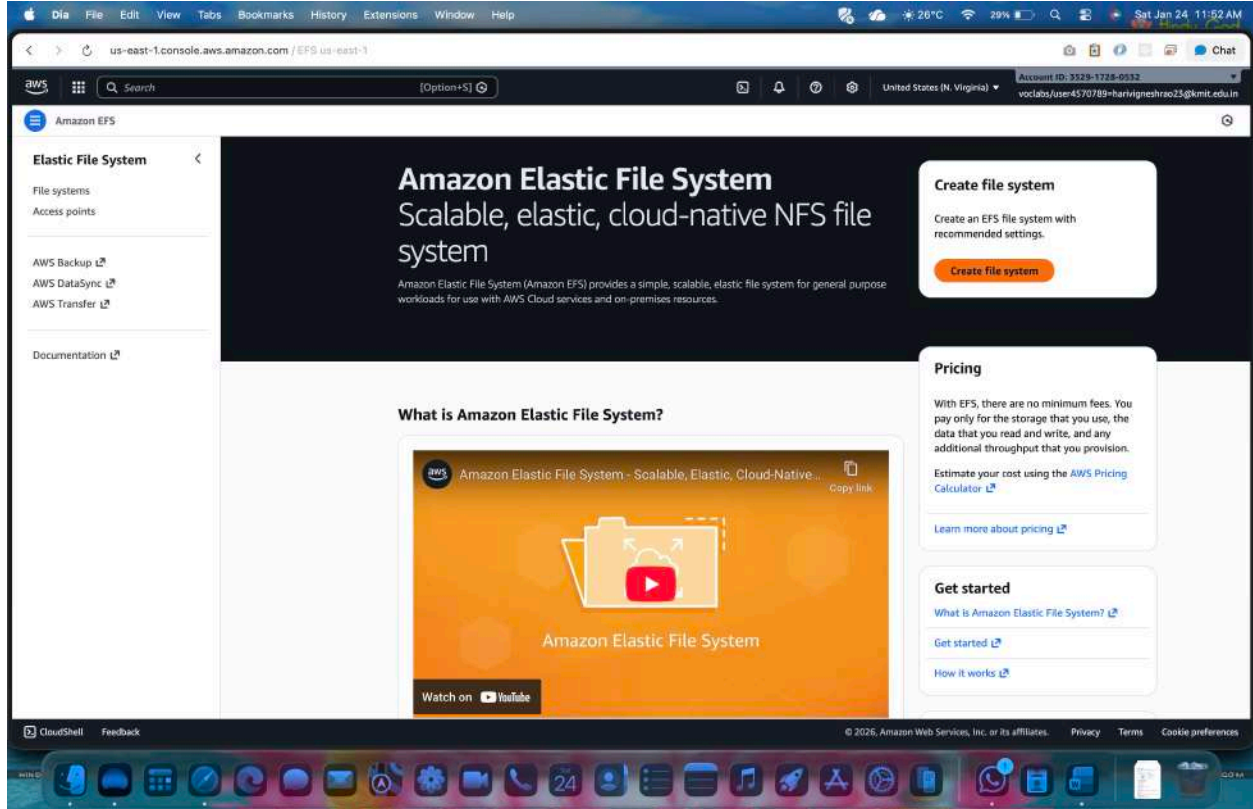


8. Repeat the above steps to launch another instance named **EFS-2** in **Subnet-1c** with the same configuration.



Step 2: Creating an EFS File System

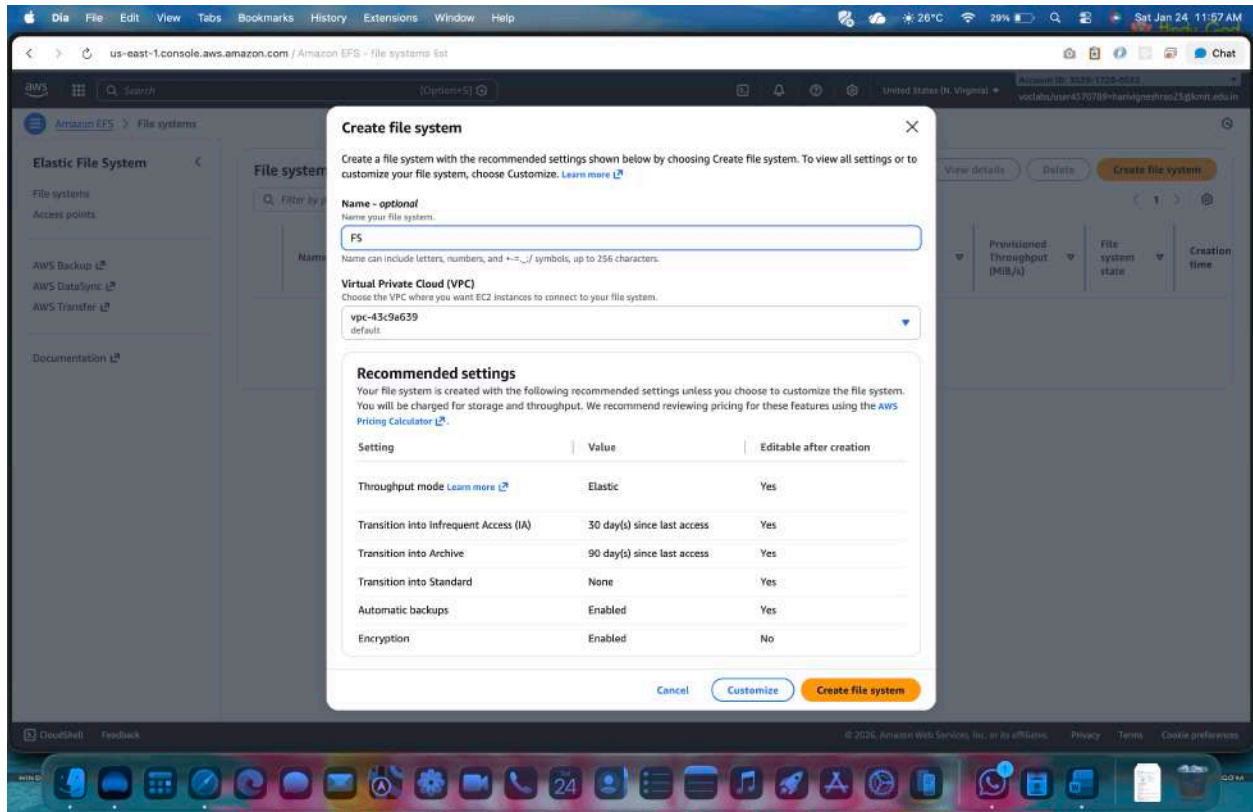
1. Go to the EFS service in the AWS Management Console.
2. Click on "Create file system".



3. Specify details:

- Name: Optional
- VPC: Default
- Enable region button.

Click on "Next".

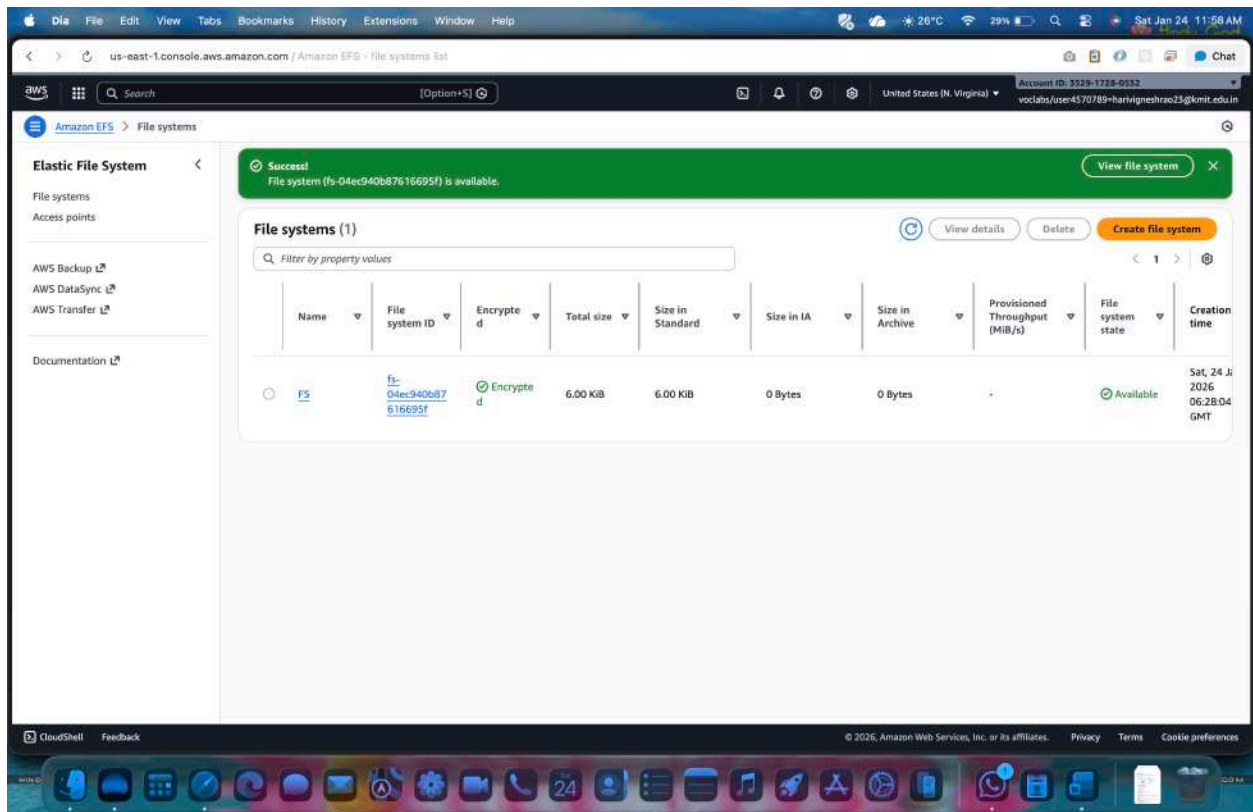
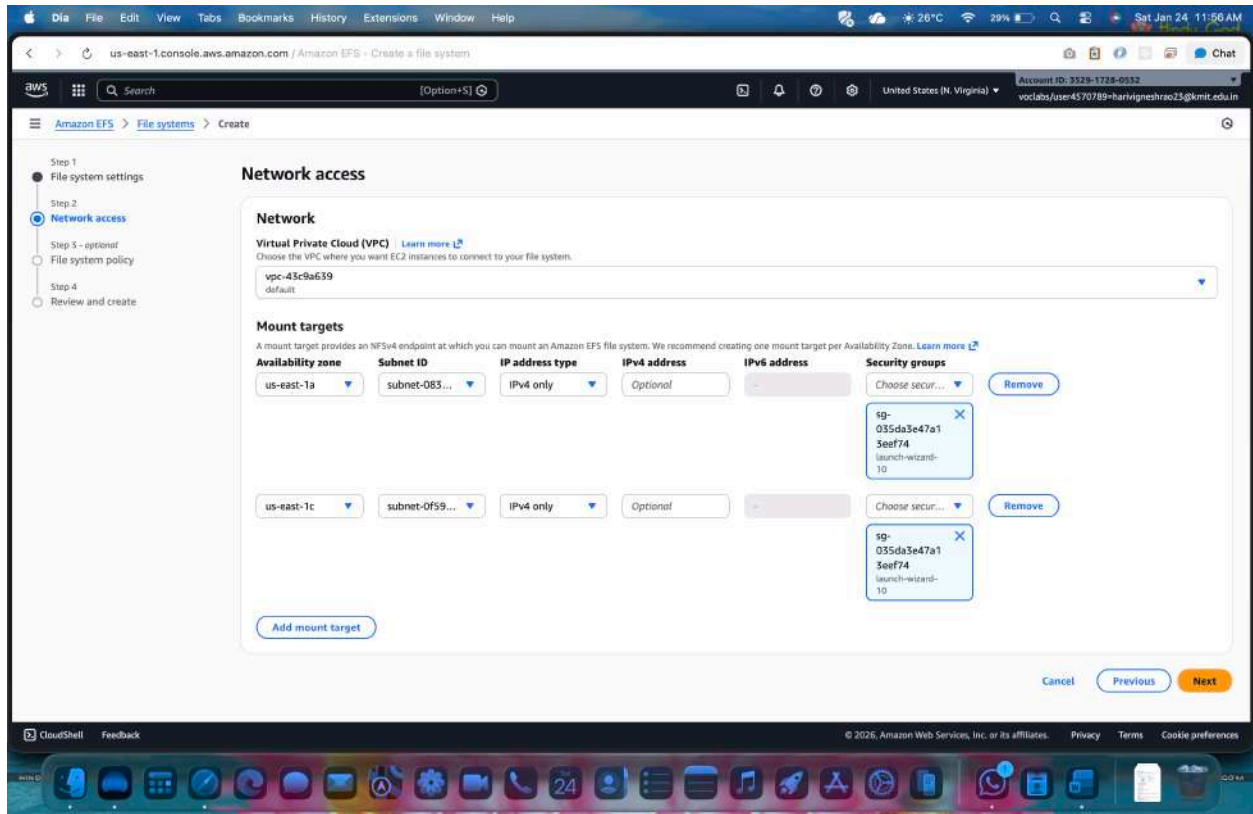


4. In the Network settings:

- Delete all security groups.
- **Select the newly created security group (NFS).**---which is created while creating instance

Click on "Next" and review the configuration.

Click on "Create file system".



Step 3: Accessing the two EC2 instances named EFS-1 & EFS -2 in **two different PowerShell** sessions and performing the specified tasks:

Accessing EFS-1 Instances in Two Different PowerShell Sessions:

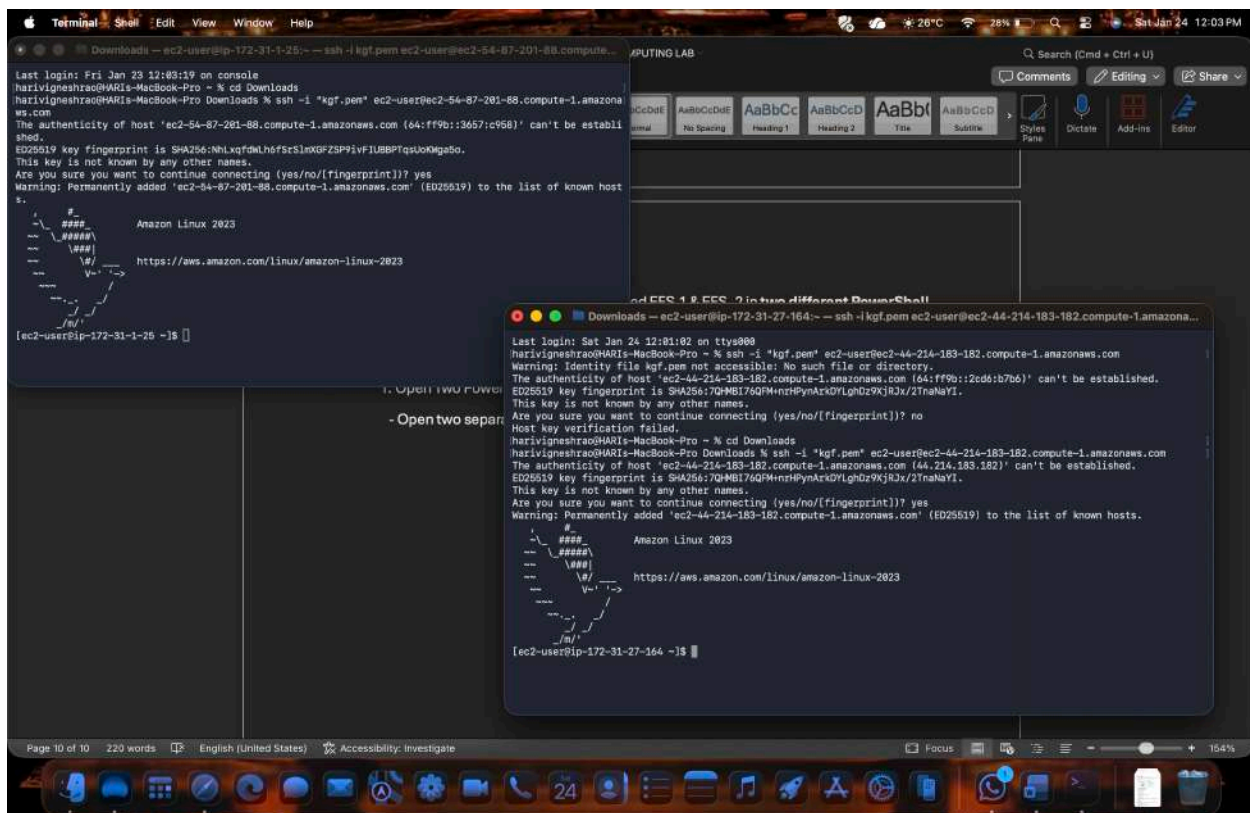
1. Open Two PowerShell Sessions:

- Open two separate PowerShell windows or tabs on your local machine.

2. SSH into the Instance:

- Use the SSH command to connect to the EFS-1 instance

`ssh -i [path-to-your-keypair.pem] ec2-user@[instance-public-ip]`



3. Switch to Root User:

- Gain root access by executing the following command:

`sudo su`

4. Create a Directory:

- Make a directory named "efs" using the following command:

mkdir efs

5. Install Amazon EFS Utilities:

- Use yum package manager to install the Amazon EFS utilities:

yum install -y amazon-efs-utils

```
Terminal Shell Edit View Window Help
Downloads -- root@ip-172-31-1-25:/home/ec2-user -- ssh -i kgt.pem ec2-user@ec2-54-87-201-88.compute-1.amazonaws.com -- 213x53

[ec2-user@ip-172-31-1-25 ~]$ sudo su
[root@ip-172-31-1-25 ec2-user]# mkdir efs
[root@ip-172-31-1-25 ec2-user]# yum install -y amazon-efs-utils
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
241 kB/s | 38 kB 00:00

Package Architecture Version Repository Size
-----
Installing:
amazon-efs-utils x86_64 2.4.1-1.amzn2023 amazonlinux 4.7 M
Installing dependencies:
stunnel x86_64 5.58-1.amzn2023.0.2 amazonlinux 156 k

Transaction Summary
-----
Install 2 Packages

Total download size: 4.9 M
Installed size: 18 M
Downloading Packages:
(1/2): stunnel-5.58-1.amzn2023.0.2.x86_64.rpm 4.8 MB/s | 156 kB 00:00
(2/2): amazon-efs-utils-2.4.1-1.amzn2023.x86_64.rpm 55 MB/s | 4.7 MB 00:00
Total 38 MB/s | 4.9 MB 00:00

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
Installing : stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
Running scriptlet: stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
Installing : amazon-efs-utils-2.4.1-1.amzn2023.x86_64 2/2
Running scriptlet: amazon-efs-utils-2.4.1-1.amzn2023.x86_64 2/2
Verifying : amazon-efs-utils-2.4.1-1.amzn2023.x86_64 3/2
Verifying : stunnel-5.58-1.amzn2023.0.2.x86_64 2/2

WARNING:
A newer release of "Amazon Linux" is available.

Available Versions:
Version 2023.10.20260120:
Run the following command to upgrade to 2023.10.20260120:
dnf upgrade --releasever=2023.10.20260120

Release notes:
https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.10.20260120.html

Installed:
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Focus 24 154%
```

6. List Files:

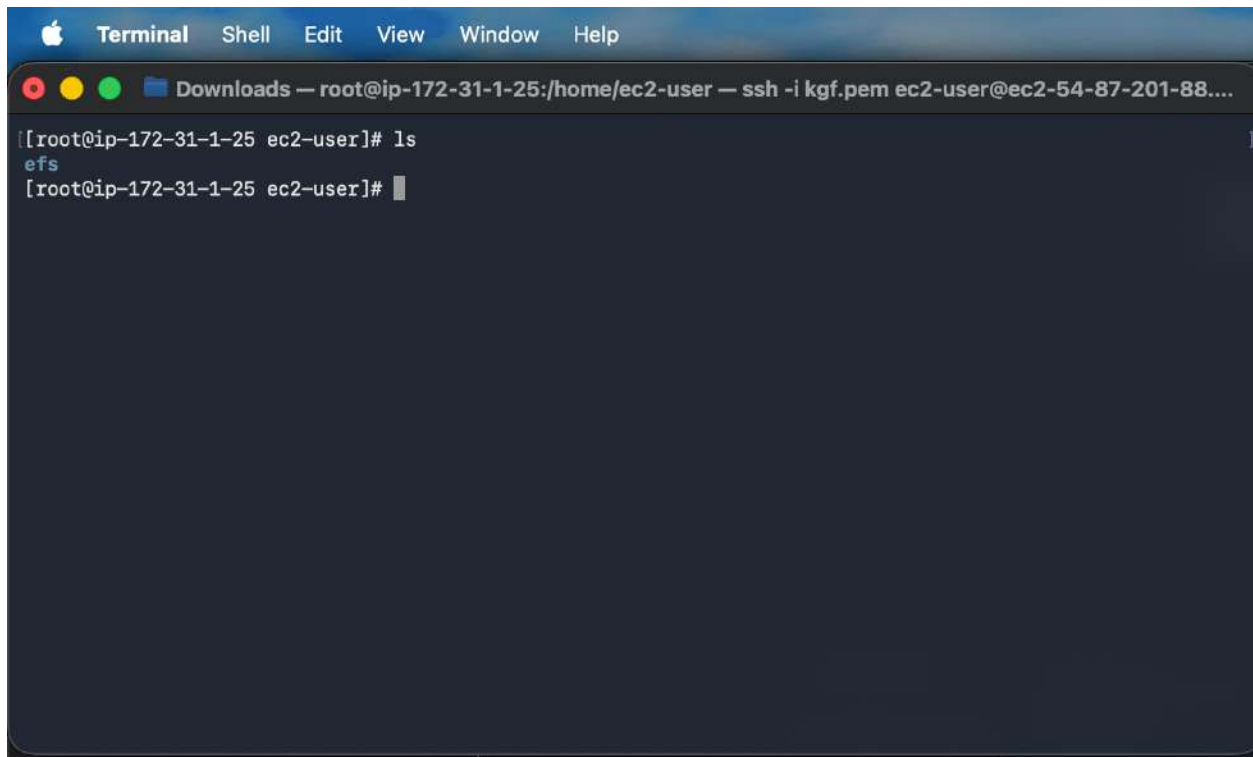
- Execute the following command to list files in the current directory:

ls

7. Verify Installation (Optional):

- Optionally, you can verify the installation of the Amazon EFS utilities by checking the version:

efs-utils --version

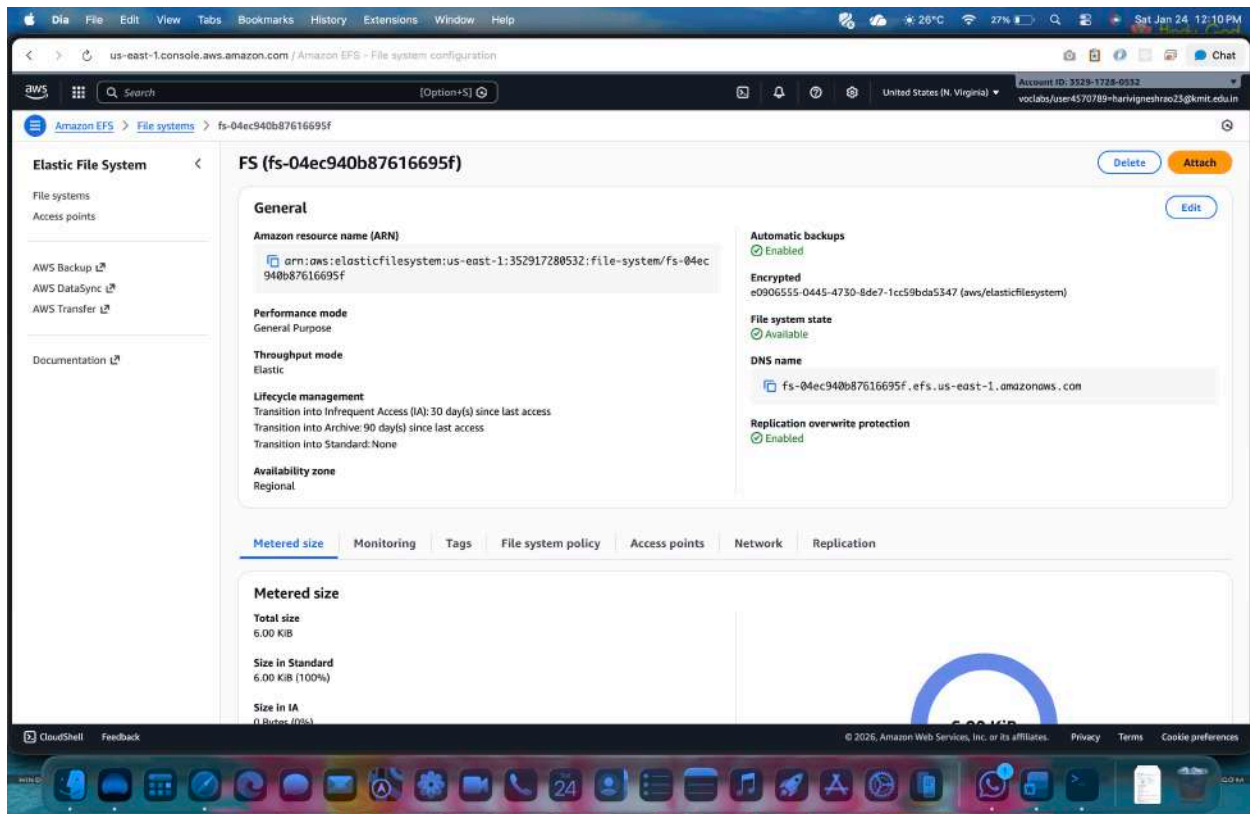
A screenshot of a macOS Terminal window. The title bar shows 'Terminal' with standard window controls. The menu bar includes 'Terminal', 'Shell', 'Edit', 'View', 'Window', and 'Help'. The window title is 'Downloads — root@ip-172-31-1-25:/home/ec2-user — ssh -i kgf.pem ec2-user@ec2-54-87-201-88....'. The terminal content shows a prompt '[root@ip-172-31-1-25 ec2-user]#', the command 'ls', and the output 'efs' in blue text. A second prompt '[root@ip-172-31-1-25 ec2-user]#' is visible below the output.

```
Terminal Shell Edit View Window Help
Downloads — root@ip-172-31-1-25:/home/ec2-user — ssh -i kgf.pem ec2-user@ec2-54-87-201-88....

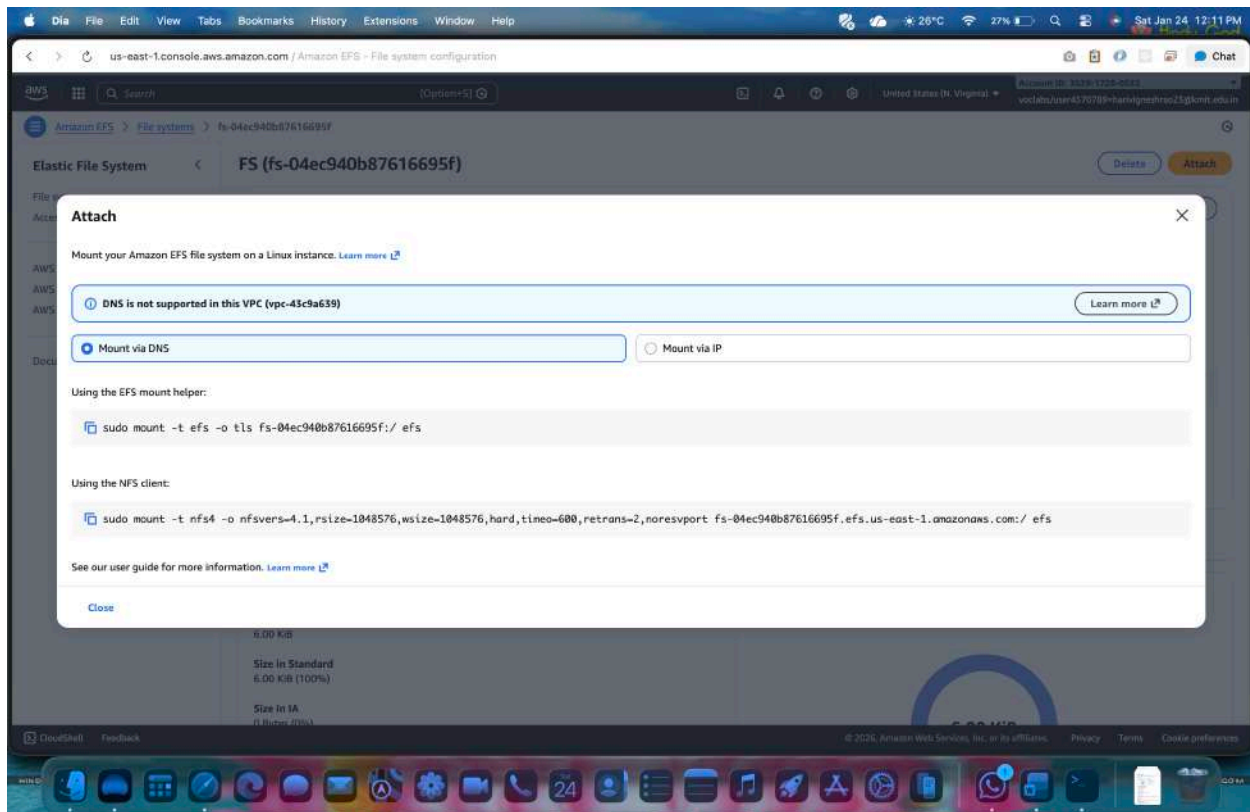
[root@ip-172-31-1-25 ec2-user]# ls
efs
[root@ip-172-31-1-25 ec2-user]#
```

Step 4: Attaching EFS to EC2 Instances

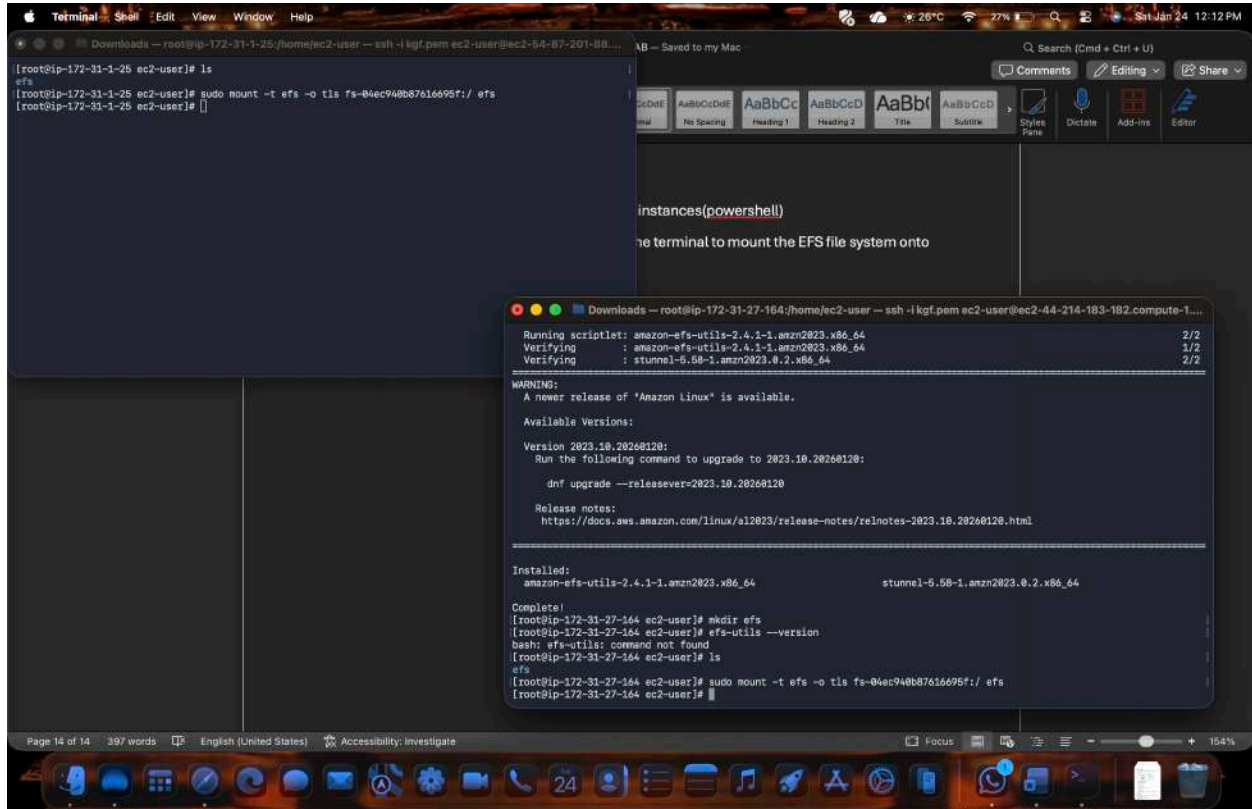
1. Go to the EFS service in the AWS Management Console.
2. Click on the target EFS file system.
3. Click on the "Attach" button.



4. Choose "Mount via DNS" option.



5. Copy the displayed command. ,on both ec2 instances(powershell)
6. Paste and execute the copied command in the terminal to mount the EFS file system onto the instance.



Step 5: Verify and Test EFS

1. Change the directory to EFS on both instances.
2. Create a file on one instance.
3. Verify that the file automatically synchronizes and appears on the other instance.

Terminal Shell Edit View Window Help

Downloads — root@ip-172-31-1-25:/home/ec2-user/efs — ssh -i kgf.pem ec2-user@ec2-54-87-201-...

```
[root@ip-172-31-1-25 ec2-user]# ls
efs
[root@ip-172-31-1-25 ec2-user]# sudo mount -t efs -o tls fs-04ec940b87616695f:/ efs
[root@ip-172-31-1-25 ec2-user]# touch fs
[root@ip-172-31-1-25 ec2-user]# ls
efs fs
[root@ip-172-31-1-25 ec2-user]# cd efs
[root@ip-172-31-1-25 efs]# touch fs
[root@ip-172-31-1-25 efs]# ls
fs
[root@ip-172-31-1-25 efs]#
[root@ip-172-31-1-25 efs]#
```

PUTTING LAB

Search (Cmd + Ctrl + U)

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3. Verify that the file auto

WARNING:
A newer release of "Amazon Linux" is available.

Available Versions:
Version 2023.10.20260120:
Run the following command to upgrade to 2023.10.20260120:
dnf upgrade --releasever=2023.10.20260120

Release notes:
<https://docs.aws.amazon.com/linux/al2023/release-notes/release-notes-2023.10.20260120.html>

Installed:
amazon-efs-utils-2.4.1-1.amzn2023.x86_64 stunnel-5.58-1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-27-164 ec2-user]# mkdir efs
[root@ip-172-31-27-164 ec2-user]# efs-utils --version
bash: efs-utils: command not found
[root@ip-172-31-27-164 ec2-user]# ls
efs
[root@ip-172-31-27-164 ec2-user]# sudo mount -t efs -o tls fs-04ec940b87616695f:/ efs
[root@ip-172-31-27-164 ec2-user]# cd efs
[root@ip-172-31-27-164 efs]# ls
fs
[root@ip-172-31-27-164 efs]#

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